

Use Case Descriptions: APC LIRA System

Use Case Name:	View Library Catalog	
Scenario:	Student or Faculty searches and browses the library's available resources.	
Triggering Event:	User accesses the LIRA portal to find academic materials or books.	
Brief Description:	This use case allows users to search, filter, and view details of library materials. It includes the backend catalog management system to fetch records and may optionally trigger the book borrowing process.	
Actors:	Student/Faculty	
Related Use Cases:	Includes: Manage Library Catalog. Extends: Borrow Books.	
Stakeholders:	Student/Faculty: To easily find required resources. Librarian: To ensure catalog visibility.	
Preconditions:	User must be connected to the LIRA system interface. The library catalog database must be online.	
Postconditions:	The requested catalog items and their details are successfully displayed to the user.	
Flow of Activities:	Actor	System
	1. Student/Faculty accesses the catalog search interface. 2. Enters search criteria (e.g., title, author, keyword). 3. Selects a specific item from the results to view details.	1.1 Display the catalog search interface. 2.1 Invoke <i>Manage Library Catalog</i> (include) to retrieve matching database records. 2.2 Display a list of matching catalog items. 3.1 Display detailed information for the selected item, including availability and location.
Exception Conditions:	2.1 If no items match the search criteria, the system displays a "No results found" message. 3.1 If the viewed item is available and the user wishes to acquire it, the user optionally invokes the <i>Borrow Books</i> (extend) use case.	

Use Case Name:	Manage Library Catalog	
Scenario:	Librarian adds, edits, or deletes records within the library database.	
Triggering Event:	A Librarian needs to update the inventory, or a user search query triggers a backend fetch.	
Brief Description:	The librarian maintains the LIRA catalog database. This process is also invoked by the system whenever catalog data needs to be retrieved for user viewing.	
Actors:	Librarian	
Related Use Cases:	Included by: View Library Catalog. Extends: Manage Borrowed Books.	
Stakeholders:	Librarian: To maintain accurate library records. System Administrators: To ensure database integrity.	
Preconditions:	Librarian must be authenticated and authorized to access the management portal.	
Postconditions:	The catalog database is successfully updated, saved, and synchronized.	
Flow of Activities:	Actor System	
	1. Librarian accesses the catalog management dashboard. 2. Selects an action to add, edit, or remove a catalog item. 3. Submits the new item details or confirms the deletion.	1.1 Display the management dashboard and current catalog list. 2.1 Display the appropriate data entry form or confirmation prompt. 3.1 Validate the submitted data. 3.2 Update the catalog database. 3.3 Display a success confirmation message.
Exception Conditions:	3.1 If data validation fails (e.g., missing ISBN), the system highlights errors and prompts the Librarian to correct them. 3.2 If modifying an item that is currently checked out, the system optionally invokes <i>Manage Borrowed Books</i> (extend) to resolve status conflicts.	

Use Case Name:	Borrow Books	
Scenario:	User checks out an available book from the library catalog.	
Triggering Event:	User decides to borrow a specific catalog item while viewing it.	
Brief Description:	Handles the checkout process for a user. It records the transaction by communicating with the Koha system to manage the borrowed status.	
Actors:	Student/Faculty (Implicit via View Library Catalog)	
Related Use Cases:	Extended from: View Library Catalog. Includes: Manage Borrowed Books.	
Stakeholders:	Student/Faculty: To successfully acquire a book. Koha: To track inventory lifecycle.	
Preconditions:	User must have an active, valid library account. The selected item must be available in inventory.	
Postconditions:	The item is marked as borrowed, the user's account reflects the loan, and a due date is assigned.	
Flow of Activities:	Actor System	
	1. User initiates a borrow request for a selected item. 2. Confirms the borrow request and accepts terms.	1.1 Verify the user's account status and current borrowing limits. 1.2 Prompt user for final confirmation. 2.1 Invoke <i>Manage Borrowed Books</i> (include) to log the transaction via Koha. 2.2 Update the item status to 'Checked Out'. 2.3 Display the borrowing confirmation receipt and due date to the user.
Exception Conditions:	1.1 If the user has a blocked account or overdue items, the system denies the borrow request and displays a warning. 2.1 If the item becomes unavailable during the transaction, the system cancels the process and notifies the user.	

Use Case Name:	Manage Borrowed Books	
Scenario:	The system records, tracks, and updates the circulation status of library materials.	
Triggering Event:	A book is borrowed by a user, or the catalog management system requires a status override.	
Brief Description:	This use case handles the backend circulation logic. It interfaces with the Koha system to maintain accurate logs of who has borrowed what, and handles due dates and returns.	
Actors:	Koha	
Related Use Cases:	Included by: Borrow Books. Extended by: Manage Library Catalog.	
Stakeholders:	Librarian: For oversight of circulating inventory. Koha System: To maintain synchronization with the central ILS.	
Preconditions:	The LIRA system must have an active connection to the Koha system API/database.	
Postconditions:	The circulation record is created or updated in both LIRA and Koha.	
Flow of Activities:	Actor System	
	1. Koha receives circulation update parameters (e.g., checkout, check-in, renew). 2. Koha confirms the transaction log is committed.	1.1 The LIRA System validates the circulation data against internal rules. 1.2 Transmit the updated status to the Koha actor. 2.1 Update the local LIRA interface to reflect the synchronized status.
Exception Conditions:	1.2 If the Koha system is unreachable, the system queues the transaction and alerts the Librarian. If invoked via <i>Manage Library Catalog</i> (extend) for a lost book, the system marks the borrowed record as resolved/lost and removes it from active circulation.	

Use Case Name:	Ask AI Chatbot	
Scenario:	A user asks the intelligent assistant a reference or research question.	
Triggering Event:	Student or Faculty opens the LIRA chatbot interface and submits a query.	
Brief Description:	Allows users to interact with an AI-driven assistant for research help, finding materials, or answering FAQs. This triggers the backend management of the chatbot to formulate a response.	
Actors:	Student/Faculty	
Related Use Cases:	Includes: Manage Chatbot.	
Stakeholders:	Student/Faculty: For immediate research assistance. Librarian: To reduce manual reference desk queries.	
Preconditions:	The user is on the LIRA interface. The AI engine is online and accessible.	
Postconditions:	The user receives a relevant, accurate response or resource recommendation from the chatbot.	
Flow of Activities:	Actor System	
	1. Student/Faculty types and submits a question into the chat widget.	1.1 Receive and parse the user's input. 1.2 Invoke <i>Manage Chatbot</i> (include) to route the query to the AI engine. 1.3 Format the retrieved answer and display it in the chat interface.
Exception Conditions:	1.2 If the AI engine cannot understand the query, the system displays a fallback message asking the user to rephrase. 1.3 If the AI service times out, the system apologizes and provides contact info for a human librarian.	

Use Case Name:	Manage Chatbot	
Scenario:	The AI processes queries, or the Librarian configures the AI's knowledge base.	
Triggering Event:	A query is routed from the "Ask AI Chatbot" use case, or a Librarian logs in to tune the AI models.	
Brief Description:	This use case encompasses both the automated generation of responses by the AI actor and the administrative tuning of the chatbot's parameters and knowledge by the Librarian.	
Actors:	Librarian, AI	
Related Use Cases:	Included by: Ask AI Chatbot.	
Stakeholders:	Librarian: To ensure the AI provides accurate and appropriate answers. AI Provider: To process tokens and return responses.	
Preconditions:	For AI: A query must be received. For Librarian: Must be authenticated as an administrator.	
Postconditions:	A response is generated for the user, or the chatbot's configuration is successfully updated.	
Flow of Activities:	Actor System	
	<i>(Automated Flow)</i> 1. AI receives the prompt and contextual data. 2. AI generates a synthesized response and returns it. <i>(Admin Flow)</i> 3. Librarian accesses the Chatbot configuration panel. 4. Librarian updates knowledge sources or custom prompts and saves.	1.1 System passes the sanitized user query to the AI actor. 2.1 System logs the prompt and response for analytics. 3.1 System displays the configuration panel. 4.1 System validates and applies the new settings to the AI engine.
Exception Conditions:	2.1 If the AI generates content violating safety policies, the system intercepts the response and substitutes a safe default message. 4.1 If the Librarian uploads an incompatible knowledge document, the system rejects it and shows an error message.	

Use Case Name:	Manage Reports	
Scenario:	Librarian generates analytics regarding library usage, catalog health, and chatbot interactions.	
Triggering Event:	Librarian requests a statistical or operational report from the system.	
Brief Description:	Allows the Librarian to compile and export data reports. It involves extracting circulation data from Koha and interaction data from the LIRA chatbot.	
Actors:	Librarian, Koha	
Related Use Cases:	None	
Stakeholders:	Library Management: To make data-driven decisions regarding acquisitions and system usage.	
Preconditions:	Librarian must have administrative privileges. Log data must exist in LIRA and Koha databases.	
Postconditions:	A formatted report is generated and available for viewing or download.	
Flow of Activities:	Actor System	
	1. Librarian selects the report type (e.g., circulation stats, AI usage) and date range. 2. Koha provides the requested circulation raw data. 3. Librarian exports the final report.	1.1 Request necessary data from the local database and the Koha actor. 2.1 Aggregate and format the combined data. 2.2 Display the report preview on the dashboard. 3.1 Generate the downloadable file (PDF/Excel) and provide it to the user.
Exception Conditions:	1.1 If Koha is offline, the system generates a partial report using only local LIRA data and displays a warning note. 2.1 If the requested date range contains no data, the system outputs an empty report with a notification.	